

Expected-Return Inference over Complete Policies

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Research draft: empirical study incomplete

Abstract

Inference over complete policies separates two sources of randomness: selection of a decision rule and stochastic evolution of the environment. We study a Gibbs distribution that weights a policy by its expected return, rather than by an exponentiated trajectory outcome. Building on established policy-inference formulations, we develop and validate a simulator-based variational implementation with consistent decisions on state revisits. A score-function estimator uses unbiased rollout returns, an independent baseline, and an exact policy-space entropy term; it needs no particle resampling or noisy exponential weights. We separate gradient correctness, variational approximation, and control performance, and count initialization and refinement within a common interaction allowance. Preliminary stochastic GridWorld experiments show that rollout allocation and initialization matter: more independent policy draws improve refinement on a medium development map, whereas insufficient exploration produces no useful control feedback on a hard map. A first matched-budget comparison gives similar control values to exploration-assisted PPO, with different execution-dependent trade-offs. These development results support further validation, not general superiority or accurate posterior coverage. The study makes the target, approximation, deployment rule, and computational cost independently testable.

1 Introduction

A successful trajectory need not identify a good decision rule. Stochastic transitions can make the same policy succeed on one attempt and fail on another. Inference-based control must therefore specify whether it favors fortunate trajectories or policies with high expected return. A second choice concerns execution: selecting an entire policy once is generally different from independently resampling its actions on every state visit.

We investigate these choices through distributions over complete stationary deterministic policies. A sampled policy fixes an action at every decision state and retains that action on revisits. The distribution weights policies by the exponential of their expected return. This starting point has direct precedent: [Wingate et al. \(2011\)](#) combine a policy prior, expected-value exponential, and policy-space Metropolis–Hastings. Information-cost approaches ([Ortega and Braun, 2011](#)) and the broader control-as-inference literature ([Levine, 2018](#)) provide related interpretations. The Gibbs distribution itself is not claimed as a new principle.

The practical question is how to approximate this distribution with a simulator while preserving its objective and evaluating its decision rule fairly. An unbiased return estimate does not become an unbiased policy weight after exponentiation. A correct invariant MCMC distribution does not ensure useful finite-run exploration. Conversely, a restricted variational family can support useful control without representing every posterior dependence.

Our implementation takes the variational route: it learns a factorized distribution over complete policies using a linear expected-return gradient and exact policy-space entropy. A separately costed PPO initializer supplies exploration, not the reference measure. The study combines an exact-gradient validation, a controlled rollout-allocation experiment, and an

execution-aware total-budget comparison. Its current empirical evidence is developmental; multi-seed held-out evaluation remains necessary.

2 Target and decision rule

Consider a finite-horizon stochastic MDP with bounded trajectory return G_H , decision states $\mathcal{D}$, and a finite class Π of stationary deterministic policies. Write $J(\pi) = \mathbb{E}[G_H \mid \pi]$. For a full-support reference μ and inverse temperature $\beta > 0$, define

$$P_\beta(\pi) = \frac{\mu(\pi)e^{\beta J(\pi)}}{Z_\beta}, \quad Z_\beta = \sum_{\pi \in \Pi} \mu(\pi)e^{\beta J(\pi)}. \quad (1)$$

We use uniform μ , so its modes maximize expected return within Π . Finite-temperature samples need not be optimal controllers. This generalized posterior describes preference-weighted policy selection, not calibrated epistemic uncertainty about unknown dynamics.

Expectation precedes exponentiation. In general, $e^{\beta \mathbb{E}[G_H \mid \pi]} \neq \mathbb{E}[e^{\beta G_H} \mid \pi]$. For example, policies returning zero surely or ± 1 equiprobably have equal expected return. Equation (1) assigns them equal mass under a uniform reference, whereas exponential trajectory weighting favors the variable policy by a factor $\cosh(\beta)$. This is a change of objective, not merely an imperfect estimate of the intended weight.

Commitment and action resampling. A draw $\pi \sim Q$ followed throughout an episode gives value $\mathbb{E}_Q[J(\pi)]$. Executing Q 's statewise action marginals afresh on every visit generally gives a different value. Even a factorized policy distribution preserves repeated-visit consistency under commitment; marginal execution does not. We report both rules, using native stochastic execution for practical comparisons with stochastic RL baselines.

Stationarity is an additional restriction: the finite-horizon optimum may depend on time-to-go. Dynamic programming over time supplies an upper control, not a same-class stationary competitor. Reward rescaling also requires reciprocal rescaling of β ; temperature is part of the target specification, not a parameter selected on held-out outcomes.

3 Direct variational policy inference

Let $q_\theta(\pi) = \prod_{s \in \mathcal{D}} q_{\theta,s}(\pi(s))$ be a categorical mean-field distribution. Its objective is

$$\mathcal{L}(\theta) = \beta \mathbb{E}_{q_\theta}[J(\pi)] - \text{KL}(q_\theta \parallel \mu), \quad (2)$$

$$\log Z_\beta - \mathcal{L}(\theta) = \text{KL}(q_\theta \parallel P_\beta). \quad (3)$$

The identity separates optimization from approximation: a well-optimized factorized family can still miss policy correlations. Under uniform μ , the negative KL equals the sum of categorical entropies minus $|\mathcal{D}| \log |\mathcal{A}|$. This is entropy over policy tables, not visitation-weighted action entropy.

Simulator gradient. Draw N independent policies and K independent rollouts per policy. Set $\bar{G}_i = K^{-1} \sum_k G_H(\pi_i, \xi_{ik})$ and, for $N > 1$, $b_i = \beta(N-1)^{-1} \sum_{j \neq i} \bar{G}_j$. Let M_{is} indicate that state s is visited by at least one of policy i 's rollouts. The implemented gradient estimator is

$$\hat{g} = \frac{1}{N} \sum_{i=1}^N (\beta \bar{G}_i - b_i) \sum_{s \in \mathcal{D}} M_{is} \nabla_\theta \log q_{\theta,s}(\pi_i(s)) - \nabla_\theta \text{KL}(q_\theta \parallel \mu). \quad (4)$$

The return and baseline are detached in the score-function surrogate (Williams, 1992); the KL term is differentiated exactly.

To justify the mask, condition on all other policy coordinates and simulator randomness. Whether s is first visited cannot depend on $\pi(s)$, which has not yet been used. On the event of no visit, the return is independent of that action, and the omitted score contribution has expectation zero. The masked score itself has zero mean. The leave-one-out baseline is independent of policy i and its rollouts, so it also contributes zero in expectation. Consequently, Equation (4) is unbiased under the declared independent mean-field sampling scheme. Each policy coordinate is scored once, not once per visit.

Because the return term is linear, $K = 1$ is valid: an unbiased rollout return suffices for an unbiased gradient. Larger K trades lower conditional rollout variance against fewer policy draws or updates at a fixed interaction budget. This argument would not justify exponentiating one noisy return as an exact inference weight.

Initialization and computation. PPO-Clip (Schulman et al., 2017) first trains a statewise guide. Its clipped, positive action probabilities initialize the logits; they do not replace μ . Adam then updates the variational logits using original environment rewards and Equation (4). Fresh policies are sampled at each update, with no persistent particle genealogy or resampling stage. Development runs use $N = 32$, $\beta = 64$, and learning rate 0.01.

The total simulator allowance includes both initialization and refinement. Before sampling a minibatch, the runner reserves its worst-case NKH transitions, reducing N or stopping when insufficient allowance remains. Returns are not selectively truncated to fit a budget. Complete-episode baseline trainers receive a stopping threshold that reserves their maximum one-episode overshoot. Core baseline updates are unchanged, and actual consumed transitions and unused allowance are reported. Exact model evaluation is offline only; it never supplies a training score or gradient.

4 Evaluation and development evidence

Correctness and approximation are different checks. On small enumerable MDPs, exact differentiation of Equation (2) is compared with the expectation of the actual implementation. Enumerating policy pairs and stochastic visitation outcomes with $K = 1, 2$ and the baseline enabled or disabled yields agreement to 10^{-12} on the diagnostic instance. This validates the estimator there, not global optimization or posterior equality on larger domains.

A separate four-policy MDP makes the family restriction explicit: three policies return 1 and the fourth returns 0. At $\beta = 64$, a certified grid search bounds the best mean-field reverse KL between 0.369 and 0.405 nats, even though its expected return and the exact target’s are both essentially 1. The thesis gives the finite-grid error bound. High control return therefore does not certify an accurate target distribution.

Alternative inference engines expose different limitations. A fixed-bank SMC screen on the medium map retained only three or four root ancestors from 256 particles and gave near-zero goal probability in three seeds. A guide-field replica-exchange screen found good policies but had return effective sample sizes of only 28–66 and maximum between-chain coordinate total variation 0.578. Neither outcome supports accurate medium-scale posterior sampling. These are diagnostic alternatives with differing costs and, for exact-score MCMC, stronger model access; they are not pooled into a simulator-efficiency ranking.

Rollout allocation. Stochastic GridWorld development maps have 8×8 , 16×16 , and 32×32 cells. A fixed medium-map guide trained for 200,015 transitions initialized three independent refinements per setting. At approximately one million additional transitions,

Table 1: Medium GridWorld development comparison: one map, training seed 572, and a 1.2M total-transition cap. Parentheses are Monte Carlo standard errors over 256 committed-policy draws, not uncertainty across training runs. Native means action resampling. Wall time includes offline evaluation.

Method	Committed (SE)	Native	Seconds
VI (PPO-initialized)	0.725 (0.044)	0.992	40.9
PPO + count bonus	0.656 (0.098)	1.015	109.9
PPO	0.008 (0.137)	0.766	103.9

$K = 1$ produced committed returns 0.581, 0.595, and 0.548, compared with -0.144, -0.244, and -0.087 for $K = 4$. The former allowed 662–669 optimizer updates, versus 157–160 for the latter. Both settings improved the paired guide value and estimated ELBO on each seed. Evaluation used 512 independent policy draws coupled across each guide/refinement pair. The common guide and map limit generalization; these are not six independent end-to-end training runs.

Full-budget attribution. Table 1 reports an end-to-end medium-map check with a fresh guide and a strict 1.2-million-transition cap. VI used nominally 200k guide transitions and the remainder for $K = 1$ refinement. The guide used count-bonus coefficient 1.0, while refinement used original rewards. We therefore report both standard PPO and a separately labeled PPO control with the same bonus coefficient throughout training. Both PPO controls started from scratch; neither resumed the VI guide optimizer.

Against assisted PPO, VI has a slightly higher committed value and a slightly lower native value in this run. The committed-value difference 0.069 is smaller than the combined conditional evaluation standard error. Local wall time favors VI here, but one seed establishes neither superiority, equivalence, nor a general speedup. A comparison with standard PPO alone would confound inference with exploration assistance.

Exploration before refinement. The analogous easy-map run improved committed value from its guide’s 3.010 to 3.318, below the nonstationary upper control 3.451. On the hard map, the 200k-transition guide and all subsequent refinement minibatches instead returned the no-goal value -38.4. The ELBO estimate increased through reduced KL without any control improvement. Full-budget assisted PPO found its first successful batch near 538k transitions and reached native goal probability 0.995. A separately specified, conditional reallocation of the same total allowance to 1M guide and 200k refinement transitions raised committed goal probability from zero to 0.700; native marginal execution reached 0.993. The corresponding assisted-PPO probabilities were 0.637 and 0.995. The large gap between execution rules rules out treating consistency itself as a control advantage. This single-seed development diagnostic motivates guide-allocation validation, not a general hard-grid success claim.

Figure 1 makes the execution-dependent comparison explicit, including the failed short-guide allocation rather than only its successful follow-up.

Confirmatory design. The planned comparison includes REINFORCE with a value baseline, PPO-Clip with Monte Carlo advantages, fixed-temperature discrete SAC, categorical CEM, and stationary-output Double Q-learning (Williams, 1992; Schulman et al., 2017; Christodoulou, 2019; Mannor et al., 2003; van Hasselt, 2010). Native execution, information access, exploration assistance, and actual total cost are reported explicitly. Hyperparameters and temperature must be frozen before held-out maps are used. Independent training runs and maps, not within-run draws, are the statistical units (Agarwal et al., 2021). Family-error and revisit diagnostics complement return and success.

Execution rule changes the comparison

Development evidence | one seed per map | 1.2M-transition cap

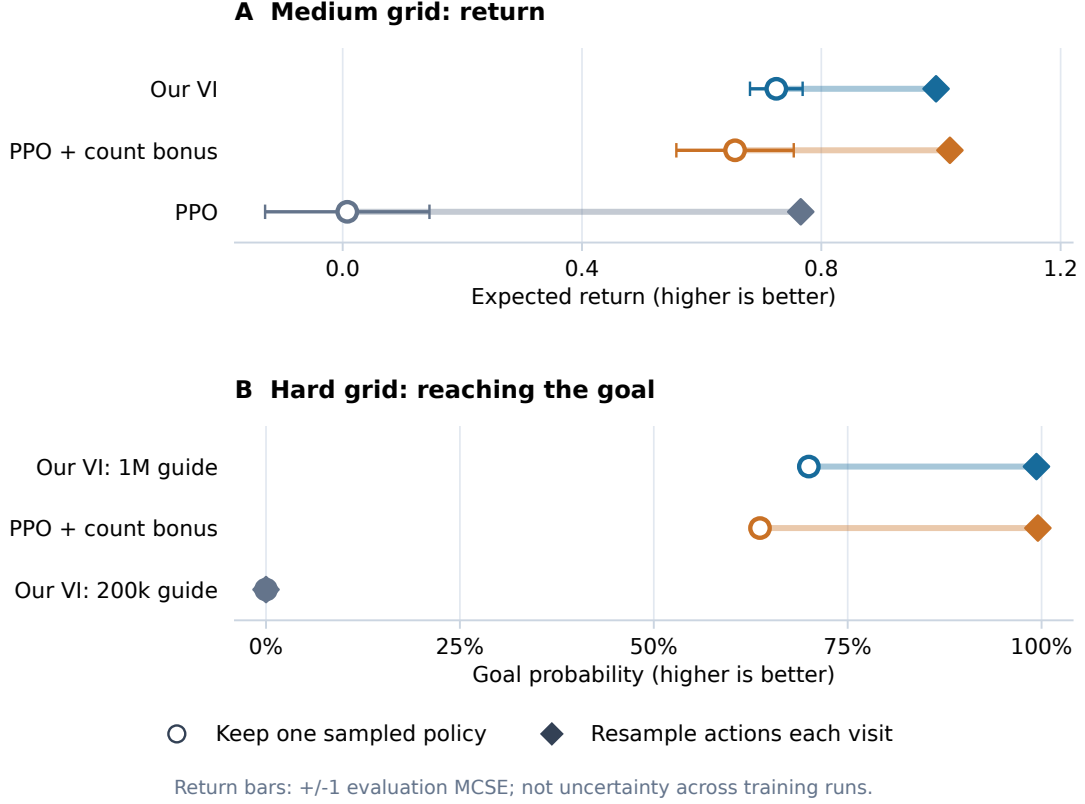


Figure 1: Execution-aware development comparison with a common 1.2M-transition cap. The medium panel uses seed 572 and the hard panel seed 573, each on map zero. Open circles commit to one sampled policy; filled diamonds resample actions on each visit. The return error bars are one conditional evaluation standard error over 256 policy draws, not training-level uncertainty. The hard-panel guide allocations retain the same total allowance. PPO with the count bonus is an explicitly assisted control; standard PPO is separate. These single-seed results motivate, but do not replace, multi-seed comparison.

5 Discussion

Expected-return inference is a target specification; it is not tied to a particular sampler. Direct variational learning avoids noisy exponential weights and permits simulator-only refinement, but introduces a restricted family and stochastic optimization. Its policy-table entropy remains an explicit preference, not a guarantee of higher control return. The present evidence identifies a useful rollout-allocation choice and a clear exploration bottleneck while retaining both unfavorable and favorable outcomes. The remaining empirical task is to determine whether the observed control–cost trade-offs persist across independent guides, maps, and baselines. The accompanying thesis retains the full estimator arguments, alternative samplers, diagnostics, and development failures.

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